

CTI

Cyber Threat Intelligence (CTI) is a cybersecurity discipline that aims to produce actionable output after processing and interpreting the data collected from multiple sources. The primary goal of CTI is to inform organizations about potential cyber threats, helping them take proactive measures to minimize damages.

CTI focuses on understanding the Techniques, Tactics, and Procedures (TTPs) of attackers. It involves collecting data from multiple sources, including Indicators of Compromise (IOCs), and processing this data to generate meaningful intelligence. By correlating this information with specific organizations, CTI helps create tailored threat intelligence that is more relevant and actionable.

The field of CTI is dynamic and constantly evolving, reflecting the continuous changes in the cybersecurity industry. This evolution is necessary to keep up with emerging threats and new attack techniques.

CTI Lifecycle



Planning and Direction

The planning and direction phase is the foundation of an effective CTI program. This phase defines the goals, target audience, and the structure for intelligence

consumption. Organizations must determine what they expect from intelligence, who will use it, and which teams will act upon it.

Key considerations include:

- **Who will use the intelligence?** Intelligence may be consumed by executives, security analysts, or a dedicated Security Operations Center (SOC) team. If a SOC team exists, intelligence can be more technical. If not, intelligence should be simplified for managerial consumption.
- **Has the organization been attacked before?** Understanding past attacks helps in shaping intelligence strategies. If previous attacks were successful, intelligence must be frequently updated and used proactively.
- **Are attacks targeting the organization or individuals?** If targeting the organization, focus should be on External Attack Surface Management (EASM) to identify and manage vulnerabilities. If individuals are targeted, Digital Risk Protection (DRP) is essential to mitigate risks such as phishing and credential leaks.
- **Are other companies in your industry exposed to the same attacks you received?** Industry-based intelligence provides insights into attacks targeting similar organizations, offering shared IOCs and details on threat actors. By leveraging industry-wide intelligence, organizations can anticipate and defend against attacks before they occur, reducing potential damage.

Information collection

The information collection stage is the part where we determine what sources we will collect data from. These resources can be both internal and external. Some of the sources we will collect data from are as follows:

- **Hacker Forums**
- **Ransomware Blogs**
- **Deep/Dark Web Forums and Bot Markets**
- **Public Sandboxes**
- **Telegram/ICQ/IRC/Discord/Twitter/Instagram/Facebook/LinkedIn**
- **Surface Web(Cybersecurity Blogs etc.)**
- **Public Research Reports**

- **File Download Sites**
- **Github/Gitlab/Bitbucket etc.**
- **Public Buckets (Amazon S3/Azure Blob etc.)**
- **Shodan/Binary Edge/Zoomeye vb.**
- **Sources that provide IOC (Alienvault, Abuse.ch, MalwareBazaar vb.)**
- **Honeypots**
- **SIEM, IDS/IPS, Firewalls**
- **Public Leak Databases**

Processing

In this stage, raw data is filtered and cleaned. False positives (incorrect or misleading data that may seem important at first glance) are removed. The data is then passed through rule sets or algorithms designed to identify relevant patterns or potential threats. Correlations between data points are checked to determine which pieces of information are truly valuable. The goal is to ensure that only the necessary and accurate data is retained for further analysis.

Analysis and Production

Once the data has been processed, it's analyzed to extract meaningful insights. This is the stage where the "intelligence" is generated—interpreting raw information into something that can be acted upon, such as identifying a security vulnerability, threat, or pattern of interest. After analysis, the output needs to be tailored depending on who will use this intelligence. For instance, a report for an executive team may be different in depth and format from one intended for technical staff.

Dissemination and Feedback

This phase involves distributing the produced intelligence to the right stakeholders or teams. Depending on the audience, the information might be sent through different channels and formats. The key part here is feedback: if the intelligence is flawed or leads to false positives (for example, if a blog subdomain from a trusted organization is flagged just because the main domain is suspicious), feedback is gathered from those who use the intelligence.

By collecting and acting on this feedback, you can improve the intelligence system. For example, if [debsite.com](#) is incorrectly flagged as suspicious while a subdomain like [some.debsite.com](#) is safe, you would need to adjust your filters, algorithms, or rules to prevent this mistake. The feedback loop ensures that the intelligence system becomes more refined and accurate over time, reducing false positives and improving efficiency.

Types of Cyber Threat Intelligence



Understanding Cyber Threat Intelligence (CTI) and Its Types

Cyber Threat Intelligence (CTI) is critical for organizations to defend against cyber threats by providing insights into potential risks, attack patterns, and adversary behaviors. It is categorized into four types: **Technical, Tactical, Operational, and Strategic**, each serving a distinct purpose within an organization's security framework.

1. Technical Cyber Threat Intelligence

Purpose: Immediate defense against known threats.

Users: SOC Analysts, Incident Responders.

Technical CTI focuses on **Indicators of Compromise (IOCs)** such as:

- Malicious IP addresses
- Phishing domains

- Hashes of malware samples

These IOCs are compiled into reports, rule sets, and automated defense mechanisms (e.g., firewall rules, SIEM alerts, IDS signatures) to protect an organization from cyber threats.

Key Takeaways:

Helps in real-time detection and mitigation of threats.

Provides actionable data for security tools.

Enables rapid response to active cyber incidents.

2. Tactical Cyber Threat Intelligence

Purpose: Understanding adversary techniques to enhance security posture.

Users: SOC Managers, Security Leadership.

Tactical CTI analyzes the **Tactics, Techniques, and Procedures (TTPs)** used by attackers. It helps security teams answer critical questions such as:

- What vulnerabilities are frequently exploited?
- What are the attacker's motives and goals?
- Which industries or regions are being targeted?

By understanding these patterns, organizations can proactively strengthen their defenses against emerging threats.

Key Takeaways:

Helps security teams anticipate and counteract attacks.

Informs risk management and defensive strategies.

Bridges the gap between technical findings and strategic decisions.

3. Operational Cyber Threat Intelligence

Purpose: Proactive threat hunting and focused investigations.

Users: Threat Hunters, Security Managers.

Operational CTI also examines attacker **TTPs**, but unlike Tactical CTI, it is more focused on **specific threats, campaigns, or adversaries**. It is used for **manual investigation, threat hunting, and active defense strategies** rather than automated detection.

For example, if an organization suspects an advanced persistent threat (APT) is targeting them, Operational CTI helps analysts track their methods and behaviors to preemptively block potential attacks.

Key Takeaways:

Enables targeted investigations of cyber threats.

Supports proactive threat hunting efforts.

Provides deeper insights into evolving cyber campaigns.

4. Strategic Cyber Threat Intelligence

Purpose: Long-term decision-making and cybersecurity investments.

Users: Executives, CISOs, Business Leaders.

Strategic CTI is high-level intelligence used by executives to **align cybersecurity with business goals**. It focuses on:

- Cybersecurity trends and forecasts.
- Threat landscapes affecting the industry.
- Risk assessment for business operations.

This intelligence helps organizations **plan budgets, make security investments, and develop policies** to enhance long-term resilience against cyber threats.

Key Takeaways:

Guides executive decision-making for cybersecurity strategy.

Helps allocate resources effectively for security improvements.

Provides insight into the broader threat landscape.

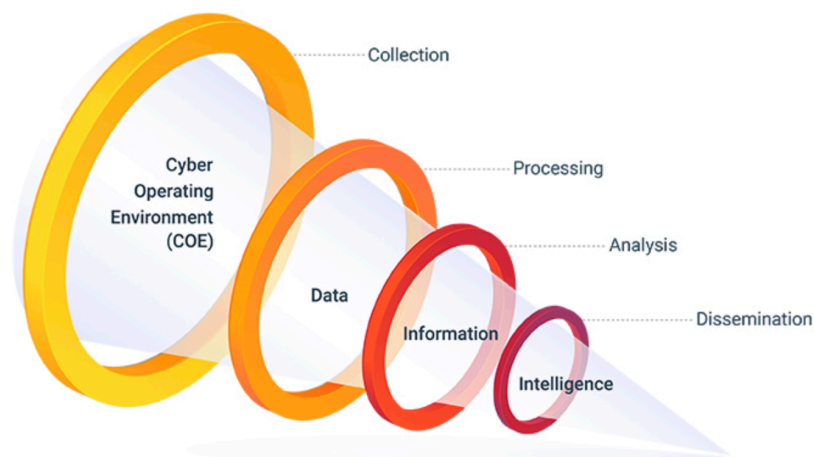
Technical CTI: Immediate threat detection and defense.

Tactical CTI: Understanding attacker methods for proactive security.

Operational CTI: Focused investigations and threat hunting.

Strategic CTI: Long-term planning and cybersecurity investment.

Interpreting Threat Intelligence Data



Threat intelligence data is often vast and complex, as it is gathered from multiple sources. If not processed correctly, it can lead to numerous false positives, reducing the quality and reliability of the intelligence. To avoid these issues, it is essential to properly understand and interpret the data before drawing conclusions.

This section will focus on the **cause-and-effect relationship in threat intelligence data interpretation** rather than delving into advanced topics like **Big Data Analysis**, which is beyond our scope. Those interested in **Big Data Analysis** can easily find relevant resources online.

Importance of Accurate Threat Intelligence Data Interpretation

Effective threat intelligence analysis requires filtering out false or misleading data to prevent unnecessary alerts and disruptions. A common example of this is when a **legitimate file hash from a trusted application, such as Microsoft software, is mistakenly flagged as malicious**. If this occurs:

- The application may be blocked within the organization.
- Critical business processes that rely on the application could be disrupted.
- Security teams may waste resources investigating a false threat instead of addressing real risks.

To **prevent such errors**, we must implement a systematic approach to analyzing and filtering intelligence data.

Key Steps in Processing Threat Intelligence Data

1. Whitelisting Legitimate Data

Before analyzing threats, it is essential to identify and filter out known safe elements. This involves:

- Creating a **whitelist** of trusted **IP addresses, domains, URLs, and file hashes**.
- Comparing newly collected data against this whitelist.
- Removing or deprioritizing alerts related to whitelisted entities to reduce false positives.

2. Classifying and Labeling Data

Threat intelligence data is often unstructured and complex. To make it more manageable, we must **classify and label** it based on its relevance to specific threat categories. This helps security teams:

- Navigate data more efficiently.
- Prioritize critical threats over less significant ones.
- Associate each data group with relevant attack surface areas.

For example, **classifying threats based on their type** could look like this:

Threat Category	Example	Associated Data Type
Malware	Ransomware campaign	File hashes, URLs
Phishing	Email spoofing	Domains, email headers
Network Intrusion	Brute force attack	IP addresses
Supply Chain Attack	Compromised vendor software	Digital certificates, software hashes

3. Associating Data with the Attack Surface

To make threat intelligence actionable, it must be mapped to an organization's **attack surface**. By correlating intelligence data with specific assets, security teams can:

- Identify which threats pose the highest risk.
- Implement targeted defensive measures.
- Continuously monitor critical systems for emerging threats.

For example:

- **If a suspicious IP address is detected** – Security teams should check whether it has interacted with company servers.
- **If a new malware hash appears in intelligence feeds** – Endpoint protection systems should be updated to detect and block it.

Interpreting threat intelligence data effectively is crucial for preventing false positives and ensuring that real threats are identified and addressed. By:

Whitelisting legitimate data to filter out unnecessary alerts.

Classifying and labeling intelligence for better organization and prioritization.

Mapping intelligence data to the attack surface to understand potential risks.

The Importance of Attack Surface in Threat Intelligence

Traditional threat intelligence models are increasingly proving inadequate in addressing modern cybersecurity challenges. The introduction of the **External Attack Surface** concept has highlighted these limitations and provided a more comprehensive approach to securing organizational assets.

A key advancement in this area is **Extended Threat Intelligence (XTI)**, which goes beyond **Classical Threat Intelligence (CTI)** by focusing on an **organization-specific attack surface**. Unlike CTI, which primarily analyzes external threats, XTI maps out an organization's **own digital footprint** to identify potential vulnerabilities.

This approach enhances visibility into assets that may have been overlooked, such as **forgotten endpoints, subdomains, or exposed services**. By understanding their attack surface, organizations can better **prioritize security measures** and protect critical assets effectively.

To accurately map an organization's attack surface, various components must be identified and analyzed. These include:

Network and Web Assets

- **Domains and Subdomains** – Ensuring all organizational web assets are accounted for.
- **Websites and Login Pages** – Identifying potential entry points for attackers.
- **CMS Applications** – Assessing vulnerabilities in content management systems.
- **Technologies Used** – Recognizing software frameworks, libraries, and third-party services.
- **IP Addresses and IP Blocks** – Mapping network infrastructure.
- **DNS Records** – Reviewing public DNS configurations for exposed data.

Employee and Corporate Data

- **C-Level Employee Emails** – Identifying publicly available executive email addresses that could be targeted in phishing attacks.

- **Network Applications** – Evaluating exposed applications and services used within the organization.
- **Operating Systems** – Understanding the OS landscape to assess potential vulnerabilities.

Financial and Security Identifiers

- **BIN Numbers and SWIFT Codes** – Identifying financial infrastructure that could be targeted in fraud or cyberattacks.
- **SSL Certificates** – Checking for misconfigured or expired certificates that may expose sensitive data.

Once raw data is analyzed in the context of an organization's attack surface, it becomes actionable intelligence. This intelligence is primarily applied in three key areas:



1. **External Attack Surface Management (EASM)**
2. **Digital Risk Protection (DRP)**
3. **Cyber Threat Intelligence (CTI)**

When combined, these three areas form the **Extended Threat Intelligence (XTI)** framework. Each component contributes uniquely to strengthening cybersecurity defenses.

1. External Attack Surface Management (EASM)

EASM focuses on continuously monitoring and managing an organization's internet-exposed assets, such as websites, domains, IP addresses, and cloud services. It helps identify vulnerabilities, misconfigurations, and unauthorized changes that could pose security risks.

Key Functions of EASM:

- Detects shadow IT assets (unregistered or forgotten digital assets).
- Monitors security vulnerabilities in externally facing infrastructure.
- Provides alerts on unauthorized changes in domain, DNS, and SSL configurations.
- Assesses the risk of exposed services and open ports.

Common EASM Alerts and Actions:

Alert Type	Recommended Action
New Digital Asset Detected	Verify ownership and ensure security controls are in place.
Domain Information Change Detected	Check WHOIS records to confirm legitimate updates.
DNS Information Change Detected	Monitor for unauthorized DNS modifications.
DNS Zone Transfer Detected	Investigate misconfigurations that could expose sensitive data.
Internal IP Address Detected	Ensure internal IPs are not publicly exposed.
Critical Open Port Detected	Close or secure unused open ports.
SMTP Open Relay Detected	Disable open relay settings to prevent email abuse.
SPF/DMARC Record Not Found	Configure email authentication to prevent spoofing.
SSL Certificate Revoked/Expired	Renew SSL certificates to maintain secure connections.
Suspicious Website Redirection	Investigate and block unauthorized redirections.

Subdomain Takeover Detected	Fix DNS misconfigurations to prevent exploitation.
Website Status Code Changed	Identify unexpected downtime or unauthorized changes.
Vulnerability Detected	Apply patches or mitigations immediately.

By proactively managing the external attack surface, organizations can minimize potential entry points for cyber attackers.

2. Digital Risk Protection (DRP)

DRP focuses on identifying and mitigating digital threats that could impact an organization's brand reputation, financial security, or supply chain. It monitors risks across the surface web, deep web, and dark web, detecting cyber threats before they materialize into attacks.

Key Functions of DRP:

- Detects phishing attempts and impersonation of brand assets.
- Identifies financial fraud and illicit activities targeting the organization.
- Assesses risks related to third-party vendors and supply chains.
- Monitors cyber threats against key executives and high-profile personnel.

Common DRP Alerts and Actions:

Alert Type	Recommended Action
Potential Phishing Domain Detected	Investigate and request takedown of fraudulent sites.
Rogue Mobile Application Detected	Identify unauthorized mobile apps and remove them.
IP Address Reputation Issue	Investigate if organization-owned IPs have been blacklisted.
Compromised Employee Credentials Found	Reset affected passwords and implement multi-factor authentication (MFA).
Leaked Sensitive Data on Dark Web	Assess data exposure and strengthen security controls.
Fake Social Media Profile Detected	Report and remove impersonation accounts.

DRP helps organizations safeguard their digital presence by proactively addressing emerging risks and preventing reputational damage.

3. Cyber Threat Intelligence (CTI)

CTI involves collecting, analyzing, and sharing information about cyber threats, including tactics, techniques, and procedures (TTPs) used by threat actors. It enables security teams to anticipate, detect, and respond to cyber threats more effectively.

Key Functions of CTI:

- Provides early warning of cyber threats.
- Enhances security incident detection and response.
- Helps prioritize vulnerability management based on threat relevance.
- Supports security teams with in-depth analysis of attack methodologies.

Common CTI Alerts and Actions:

Alert Type	Recommended Action
Malware Campaign Targeting the Industry	Strengthen endpoint protection and apply threat intelligence feeds.
Zero-Day Vulnerability Identified	Implement temporary mitigations until a patch is available.
Suspicious IP Address Activity	Block malicious IPs and monitor network traffic.
New Ransomware Variant Emerging	Update security tools with latest ransomware detection rules.
Dark Web Discussion About Organization	Investigate and assess potential threats or leaked data.

CTI empowers organizations to stay ahead of cyber threats by integrating real-time intelligence into their security strategies.

By leveraging **EASM, DRP, and CTI**, organizations can proactively manage their attack surface, protect their digital assets, and enhance their cybersecurity resilience. These components form the **Extended Threat Intelligence (XTI)** framework, which provides a comprehensive approach to cybersecurity by:

Identifying and mitigating vulnerabilities in external assets.

Protecting the organization's digital presence from brand impersonation and fraud.

Gaining actionable insights into emerging cyber threats.

SOC and Threat Intelligence Integration



-Improving SIEM Efficiency

SIEM solutions aggregate and analyze security logs from various sources. However, they often generate **false positives** due to the sheer volume of data collected. **Integrating threat intelligence feeds with SIEM** helps:

- **Filter out irrelevant data** by cross-referencing logs with known threat indicators.
- **Enhance event correlation** to identify real security threats more accurately.
- **Reduce alert fatigue** for SOC analysts by minimizing false positives.

-Enhancing SOAR Automation

SOAR platforms automate and coordinate responses to security incidents. By integrating TI:

- **Automated playbooks** can act on **verified threats** instead of false alarms.
- **Faster incident response** by dynamically updating response protocols based on real-time threat intelligence.
- **Better prioritization** of alerts based on **current threat landscapes**.

-Strengthening EDR Capabilities

EDR solutions focus on detecting and responding to threats on **end-user devices**. When fed with **real-time threat intelligence**, EDR systems can:

- **Improve detection accuracy** by correlating endpoint activities with known malicious behavior.
- **Identify new attack patterns** before they cause widespread damage.
- **Mitigate risks proactively** by flagging suspicious user behavior based on global threat intelligence feeds.

-Maximizing Firewall Protection

Firewalls act as the **first line of defense** against external threats. By integrating threat intelligence with firewall configurations, organizations can:

- **Block malicious IP addresses** before they establish a connection.
- **Prevent unauthorized access** by continuously updating firewall rules with real-time threat intelligence.
- **Reduce attack surface** by proactively filtering **suspicious inbound and outbound traffic**.

For instance, if threat intelligence identifies a **malicious IP**, the firewall can **immediately block** any incoming traffic from that IP—**eliminating threats before they trigger alerts within SOC tools**.

Integrating **Threat Intelligence** with **SOC tools** enhances overall **cybersecurity posture** by:

Reducing false positives in SIEM through **intelligent data filtering**.

Automating better incident responses in SOAR based on real-time threat feeds.

Detecting endpoint risks more effectively with intelligence-driven EDR.

Blocking malicious traffic preemptively by feeding threat intelligence to firewalls.